Higher National Diploma in Engineering (Civil/Electrical/Mechanical), Higher National Diploma In Building Service Engineering, and Higher National Diploma in Quantity Surveying

Second Year, First Semester Examination-2021

IT2102-Information Technology II A

Answer only 4 questions **Time: 2 hours**

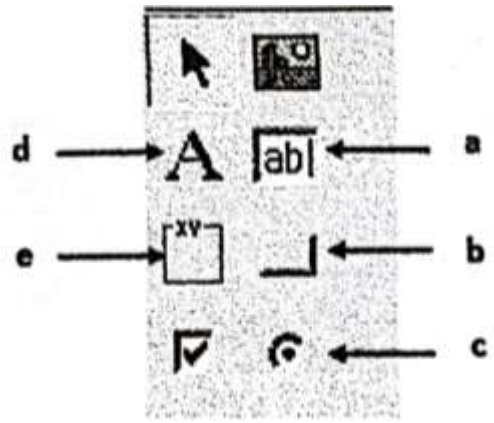
---

1. Questions

i.What is a computer programming language? Give two examples. (03 Marks)

ii.What is the purpose of a compiler/interpreter? (02 Marks)

iii.Visual Basic is an event driven programming language. Write any five events. (05 Marks)

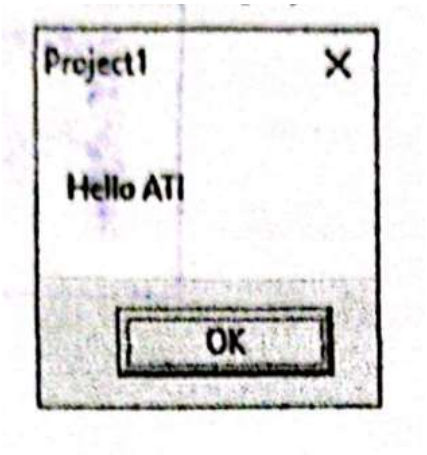


iv.Identify following controls (a, b, c, d, e) in the toolbox of VB 6.0:

v.Describe the following components of IDE of Visual Basic:

a. Property Window
b. Tool Box
c. Project Explorer Window (2 Marks each)

vi.You need to display "Hello ATI" on a message box when the form is loaded. Write the event and code to display the bellow output . (04 Marks)



---

2. Questions

i.What do you mean by variables in VB 6.0? (02 Marks)

ii.Identify valid and invalid identifiers (variable names):

a. saman b. Dim c. Integer

d. It6 e. Abc.com f. 2345 (06 Marks)

iii.Declare variables/constants to store values in the table

Value
Saman
3.14
Rs. 30.50
46.8974
GALEE2020F0001

iv.Briefly explain what are the global and local variables, using an example. (04 Marks)

v.Write the output of the following two programs:

a. VB.Net

```
Private Sub cmdBackColor_Click()
Form1.BackColor = vbRed
End Sub
```

b. VB.Net

```
Private Sub cmdMOD_Click()
Print 5 Mod 3
End Sub
```

(08 Marks)

---

3. Questions

i.What do you mean by the control structures? Give three examples for that. (05 Marks)

ii. Write the output of the following programs (06 Marks)

a.

```
Dim x As Integer
x = 10
Do While (x < 5)
Print x
x = x + 1
Loop
```

b.

```
Dim y As Integer
y = 10
Do
Print y
y = y + 1
Loop Until(y > 15)
```

c.

```
Dim z, x As Integer
z = 10
x = 1
if (x > 10) then
Print z
Else
Print z ^ 2
End IF
```

iii. Following program shows grades of a subject according to the given marks:

If marks of a subject is >=75 - "**A**" >=65 - "**B**" >=55 - "**C**", >=40 - "**S**" <40 - "**F**"

a. Design (draw) an interface to display grade of a subject according to the entered marks. You need to display output on a label and name all controls used. (04 Marks)
b. Write the code for above interface. (04 Marks)

iv. Using a loop, write a program to display sum of bellow series (0, 2, 4, 6, 8, 10, 12, 14). (04 Marks)

v.Write a program to display the bellow pattern:

*****

****

***

**

*

(03 Marks)

---

4. Questions

i.What is an array? (02 Marks)

ii. Using an array:

a. Write a program to store following names of the students, (Kumar, Saman, Arun, Nadeesha, Anne). (04 Marks)
b. Using a loop, print all names (in part a.). (04 Marks)

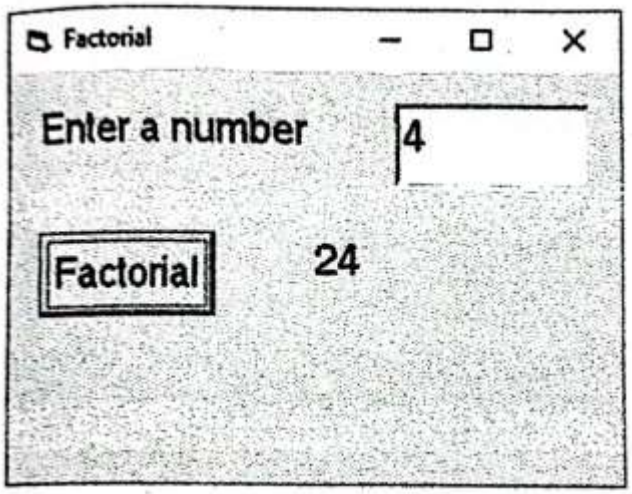
iii. The following table shows marks obtained by the students. Using a 2D array store all names and marks:

Kapila	Nuwan	Akram
78	45	60

(04 Marks)

iv. Using loops print all names with relevant marks as shown in the table. (04 Marks)

v.Write a program to find the factorial of any given positive integer for the following interface (names of the controls are TextBox-txtN, Command button - cmdFact, label - lblOut)



(06 Marks)

---

5. Questions

i.Compare User Defined Functions and Event procedure. (04 Marks)

ii. Write syntax of User Define Function. (04 Marks)

iii. Write User Define Functions for to find:

a. Area of a circle (03 Marks)
b. Volume of a cylinder (03 Marks)

iv. Design an interface to find volume of a cylinder. (03 Marks)

v.Write all controls used (in question iv). (04 Marks)

vi. Call the function written in above question **iii b** with relevant values in an event to display the output **(volume of a cylinder)** on a label. (04 Marks)